

Beyond English-Only GRPO: Training Language and Auxiliary Reward as Implicit Regularizers in Sub-3B Mathematical Reasoning

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Abstract

Background. Group Relative Policy Optimization (GRPO) drives recent advances in mathematical reasoning for sub-3B language models, but recent reproducibility analysis [1] shows that small-model RL gains may not survive multi-seed evaluation. The role of training-language choice and auxiliary reward components remains under-studied at the deployment-relevant single-GPU LoRA scale.

Methods. On the DeepSeek-R1-Distill-Qwen-1.5B base, we run a controlled multi-seed GRPO study under a single A100 80 GB LoRA constraint (rank $r=16$). Three primary arms vary exactly one axis: A1 (English training data, accuracy and format rewards), A2 (Vietnamese-translated training data, identical rewards), and A3 (English training data, with an additional fastText-based language-consistency reward R_5). Each arm is trained with three random seeds. A three-seed constant-bias ablation A4 tests whether the auxiliary-reward effect reduces to reward magnitude. Models are evaluated on AMC-23, MATH-500, AIME-2024, and 10-language MGSM.

Results. We report three orthogonal findings. **(i) Positive:** on AIME-2024 maj@8, the hardest benchmark in our suite, A3 achieves the highest mean accuracy ($37.8 \pm 1.9\%$) and the only positive effect over the untrained base (+4.4 pp), robust across three seeds. **(ii) Methodological:** A1 exhibits $\sigma = 11.3$ pp seed variance on AMC-23 and consistently degrades AIME-2024 pass@1 (mean -12.2 pp across 3/3 seeds), confirming that single-seed claims at sub-3B + LoRA scale are unreliable. **(iii) Negative:** on the MGSM 10-language benchmark, all four training conditions converge to within ± 0.5 pp of the untrained base mean, indicating that training-language choice does not produce cross-lingual transfer at this scale. The three-seed A4 ablation *fails to* reproduce A3’s effect on AIME-2024 maj@8 ($A3 - A4 = +5.58$ pp, 95% bootstrap CI $[+1.13, +11.10]$, excluding 0), refuting a pure reward-magnitude interpretation.

Conclusion. The auxiliary language-consistency reward R_5 improves the hardest-benchmark capability while preserving multilingual performance, but the mechanism is not purely a reward-magnitude perturbation. Multi-seed evaluation reveals that vanilla English GRPO at sub-3B + LoRA scale is unreliable; reproducibility studies are essential before scaling claims. We release all code, configurations, LoRA adapters, evaluation JSONs, and per-step reward traces to support replication.

Index Terms

Reinforcement learning from rewards, Group Relative Policy Optimization (GRPO), large language models, mathematical reasoning, low-rank adaptation (LoRA), cross-lingual transfer, auxiliary rewards, reward shaping, regularization, multilingual evaluation.

I. INTRODUCTION

REINFORCEMENT learning from verifier signals has become the dominant paradigm for improving mathematical reasoning in mid-sized language models. *Group Relative Policy Optimization* (GRPO) [2], [3] is the workhorse of this paradigm. By eliciting a group of candidate completions per prompt and assigning advantages relative to the group mean, GRPO removes the need for a learned value head while preserving on-policy PPO-style updates [4]. Recent work on small reasoning models (Open-RS [5]) has shown that GRPO can lift a 1.5B distilled checkpoint to 80% pass@1 on the AMC-23 competition benchmark with only 50 optimization steps and 7K training problems on four NVIDIA A40 GPUs.

Manuscript prepared 2026. Code, configurations, LoRA adapters, and per-step reward traces are publicly archived at <https://github.com/vudang4494/xling-grpo-sub3b> (Apache-2.0; Zenodo DOI: [10.5281/zenodo.20061328](https://doi.org/10.5281/zenodo.20061328)).

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Two structural assumptions, however, are baked into nearly every published GRPO recipe:

- 1) *Monolingual English training corpus*. The training problems and their reference solutions are in English; multilingual generalization is assumed rather than measured.
- 2) *Reward sum dominated by accuracy and format*. Auxiliary rewards that do not directly score correctness are rarely included; when they are, their effect is interpreted as a content signal.

The downstream behaviour of GRPO when these assumptions are perturbed has received limited systematic attention at the sub-3B scale that is most relevant to deployed systems and to independent researchers operating on commodity cloud budgets.

A. Research questions

This article addresses two questions, both targeted at the single-GPU sub-3B regime:

- Q1.** How does the choice of training language (English vs. Vietnamese-translated, holding the reward set fixed) affect downstream mathematical reasoning, and is the effect uniform across benchmarks of varying difficulty?
- Q2.** When an auxiliary reward component is added that fires nearly uniformly at 1.0 across training generations – so it carries *no* content signal – does its presence still affect the trained policy? If so, through what mechanism?

B. Contributions

We make four contributions:

- 1) Multi-seed empirical evidence (3 seeds $\times$ 4 arms) that single-seed GRPO results at sub-3B + LoRA scale are unreliable: vanilla English GRPO exhibits $\sigma=11.3$ pp variance on AMC-23 (Section VI-B).
- 2) Documentation of A3’s positive effect on the hardest benchmark in our suite: $+4.4 \pm 1.9$ pp on AIME-2024 maj@8 over the untrained base, robust across three seeds (Section VI-D).
- 3) A three-seed constant-bias ablation A4 that *fails* to reproduce A3’s AIME-2024 maj@8 result ($A3-A4=+5.58$ pp, 95% bootstrap CI $[+1.13, +11.10]$, excluding 0), demonstrating that R_5 ’s contribution is content-specific rather than a pure reward-magnitude perturbation (Section VII).
- 4) A 10-language MGSM evaluation showing that the GRPO recipe at this scale *preserves but does not improve* multilingual capability regardless of training-language choice (Section VI).

Main empirical claim. Multi-seed evaluation reveals three orthogonal patterns. **(i)** The language-consistency reward arm A3 achieves the highest mean AIME-2024 maj@8 ($+4.4 \pm 1.9$ pp over base across three seeds). **(ii)** Vanilla English GRPO A1 exhibits $\sigma=11.3$ pp seed variance on AMC-23 and consistently degrades AIME-2024 pass@1 by -12.2 ± 7.7 pp. **(iii)** On 10-language MGSM, all training conditions converge to within ± 0.5 pp of the untrained base. A three-seed constant-bias control A4 *fails* to reproduce A3’s effect on AIME-2024 maj@8 ($A3-A4=+5.58$ pp, 95% bootstrap CI $[+1.13, +11.10]$), refuting a pure reward-magnitude explanation.

C. Article structure

Section II reviews GRPO and cross-lingual reasoning literature. Section III formalizes the GRPO objective and the PPO-clipping geometry that we later invoke. Section IV specifies the three arms, reward functions, and training pipeline. Section V details the experimental setup and evaluation protocol. Section VI presents the empirical results and three findings. Section VII discusses the proposed regularization mechanism. Section VIII enumerates limitations and threats to validity. Section IX concludes.

II. RELATED WORK

A. GRPO and small reasoning models

DeepSeekMath [2] introduced GRPO, replacing PPO’s learned value head with a group-relative baseline. DeepSeek-R1 [3] scaled this technique to produce strong reasoning checkpoints whose distilled variants – including the 1.5B model used here – have become standard sub-3B baselines. Open-RS [5] explored what hyperparameter recipes are needed to lift a distilled 1.5B to 80% on AMC-23; we adopt their RS2 setup verbatim for our A1 arm. Variance-reduced and length-normalized variants (Dr.GRPO [6], DAPO [7]) are orthogonal to the axes we vary.

B. Cross-lingual mathematical reasoning

Multilingual mathematical reasoning has been studied primarily at 7 B and above. LinguaLIFT [8] proposes a two-stage instruction tuning framework. MAPO [9] and MathOctopus [10] fine-tune open models for multilingual mathematical reasoning at 7 B scale. Crosslingual reasoning under test-time scaling [11] examines inference-time multilinguality. The multilingual GSM benchmark [12] is the de-facto evaluation harness for language transfer; OpenMathInstruct-2 [13] provides English-only training data. Cross-lingual collapse [14] is the closest concurrent work, observing language drift during GRPO at ≥ 3 B scale; our setting differs in three respects: (i) sub-3 B with a deliberate single-GPU LoRA budget at which prior baselines underperform their reported numbers, (ii) controlled *training-language* ablation in addition to test-language variation, and (iii) explicit reward intervention as a treatment. M-Thinker [15] is the closest scale-matched work but evaluates only on Chinese; our setup admits direct extension to other languages without methodological change.

C. Auxiliary rewards and reward shaping

Reward shaping is well-studied in classical RL. Ng et al. [16] prove that potential-based shaping preserves the optimal policy, while Skalse et al. [17] characterize how mis-specified auxiliary rewards can induce reward hacking. Zheng et al. [18] study learned intrinsic rewards in policy gradients and use constant-bias baselines that are conceptually analogous to our A4 ablation. In language-model fine-tuning, auxiliary reward components have been used to steer models toward desired stylistic or safety properties; the conventional interpretation is that they inject content-level signal into the policy gradient. We are not aware of prior work that examines what happens when an auxiliary reward fires nearly uniformly across training samples and therefore carries minimal content information. Our A3 arm provides a natural setting to test this regime, and our hypothesis (Section VII) makes a falsifiable prediction about why such a “trivially-satisfied” auxiliary reward can still substantially affect the trained policy.

III. PRELIMINARIES: GRPO AND PPO CLIPPING GEOMETRY

We restate the relevant pieces of the GRPO objective to fix notation for the mechanistic discussion in Section VII.

A. GRPO objective

Let π_θ denote the policy, $\pi_{\theta_{\text{old}}}$ the behaviour policy at the start of the optimization step, and $\mathbf{p}$ a prompt sampled from the training distribution. For each prompt, GRPO samples a group of G completions $\{\mathbf{c}^{(1)}, \dots, \mathbf{c}^{(G)}\}$ and computes a scalar reward $R(\mathbf{p}, \mathbf{c}^{(g)}) \in \mathbb{R}_{\geq 0}$ per completion. The group-relative advantage is

$$A^{(g)} = \frac{R(\mathbf{p}, \mathbf{c}^{(g)}) - \bar{R}_{\mathbf{p}}}{\sigma_{R, \mathbf{p}} + \varepsilon}, \quad (1)$$

where $\bar{R}_{\mathbf{p}} = \frac{1}{G} \sum_g R(\mathbf{p}, \mathbf{c}^{(g)})$ and $\sigma_{R, \mathbf{p}}$ is the within-group standard deviation. Per-token importance ratios

$$\rho_t^{(g)} = \frac{\pi_\theta(c_t^{(g)} \mid \mathbf{p}, \mathbf{c}_{<t}^{(g)})}{\pi_{\theta_{\text{old}}}(c_t^{(g)} \mid \mathbf{p}, \mathbf{c}_{<t}^{(g)})} \quad (2)$$

are clipped via the standard PPO-style trust region, and the GRPO objective is

$$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}_{\mathbf{p}} \left[\sum_g \sum_t \min(\rho_t^{(g)} A^{(g)}, \text{clip}(\rho_t^{(g)}, 1 - \epsilon, 1 + \epsilon) A^{(g)}) \right] - \beta_{\text{KL}} \text{KL}(\pi_\theta \parallel \pi_{\theta_{\text{old}}}). \quad (3)$$

B. Reward decomposition with multiple components

With multiple reward components $R_1, R_2, \dots, R_K$ and weights w_k , the per-completion reward is $R(\mathbf{p}, \mathbf{c}) = \sum_k w_k R_k(\mathbf{p}, \mathbf{c})$. A useful identity for our discussion: for any constant $b \in \mathbb{R}$,

$$A^{(g)}[R + b] = \frac{(R + b) - (\bar{R} + b)}{\sigma_R + \varepsilon} = A^{(g)}[R], \quad (4)$$

i.e., adding a constant offset to every completion’s reward has no effect on the advantage. This is the textbook reason “trivial” reward components are believed to be inert. The point of Section VII is that (4) ignores the rest of the optimization machinery – in particular the clipping ratio $\rho_t^{(g)}$ and the KL penalty – which depend on *realized rewards*, not advantages, and are therefore *not* invariant to additive offsets.

IV. METHOD

A. Base model and training pipeline

All three arms start from the publicly-released `deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B` checkpoint [3]. We do *not* use a separate supervised fine-tuning stage (matching Open-RS): training is GRPO directly from the distilled BASE for 50 steps, producing one LoRA adapter per (arm, seed) cell.

B. Reward functions

All arms use two rewards from the Open-RS RS2 setup:

- R_1 *correctness*: indicator of Math-Verify [19] equivalence between the extracted answer $\hat{y}$ and the gold y^* . Extraction priority: `<answer>` tag, then `\boxed{}`, then last-number fallback.
- R_2 *format*: indicator that the completion matches a regular expression requiring both a `<think>...</think>` block and an `<answer>...</answer>` block, in that order; matches the training system prompt template.

Arm A3 adds the language-consistency reward R_5 , computed with the fastText [20] language identifier:

$$R_5(\mathbf{p}, \mathbf{c}) = \begin{cases} 0 & \text{if } |\mathbf{c}| < 10, \\ \mathbf{1}[\text{lang}(\mathbf{c}) = \text{lang}(\mathbf{p})] & \text{otherwise,} \end{cases} \quad (5)$$

where $\text{lang}(\cdot)$ is the fastText `lid.176.bin` language identifier and the $|\mathbf{c}| < 10$ guard accounts for fastText’s unreliability on very short strings. Crucially, when training data is English the model rarely code-switches; thus $R_5 \approx 1$ for almost every training generation, and R_5 provides effectively zero content-discrimination signal in this regime.

C. Three arms

English-only: training data `knoveleng/open-rs` (7 K English problems); rewards $R_1 + R_2$.
 Vietnamese-translated: training data `5CD-AI/Vietnamese-MetaMathQA-40K-gg-translated` (5 203 records extracted from a 7 K random subset by retaining only those whose response contains the Vietnamese answer marker “Đáp án là”); rewards $R_1 + R_2$.
 English with R_5 : same training data as A1; rewards $R_1 + R_2 + R_5$ with R_5 ’s “no_penalty_for_en” flag disabled so it can fire on every prompt regardless of input language.

D. Compute and hyperparameters

Hardware: 1× NVIDIA A100-SXM4-80 GB on commodity cloud infrastructure. LoRA [21]: $r=16$, $\alpha=32$, dropout 0.05, `bias=none`, target modules $\{q, k, v, o\}_{\text{proj}}$. GRPO: $G=6$ rollouts per prompt, `max_completion_length= 3584` tokens, `temperature= 0.7`, $\beta_{\text{KL}} = 0.04$, learning rate 10^{-6} , scheduler `cosine_with_min_lr` with `min_lr_rate= 0.1` and `warmup_ratio= 0.1`. Effective batch is 96 prompts (per-device batch $6 \times$ gradient accumulation 16), `bf16`, flash-attention 2, gradient checkpointing (`use_reentrant=False`). Training uses TRL [22] 0.15.2 with in-process vLLM [23] 0.7.2 rollout (`gpu_memory_utilization= 0.25`, leaving ~ 60 GB for the training pass). We save at step 50 (Open-RS RS2 peak).

E. Evaluation

We use the Open-RS `lighteval MATH_QUERY_TEMPLATE` verbatim (no system prompt; the evaluation template is embedded in the user message), with greedy decoding ($T=0$), `max_tokens= 8192`, and no early stop tokens. Aligning to this template closes the AIME-2024 gap to Open-RS reported numbers from -10 pp to -2.1 pp. Three benchmarks are used: AMC-23 [24] (40 problems, pass@1), MATH-500 [25] (500 problems, pass@1), and AIME-2024 [26] (30 problems, pass@1 plus maj@8 with seeds 42–49 to cover the standard 8-sample majority-vote evaluation).

TABLE I

MULTI-SEED RESULTS ACROSS THREE RANDOM SEEDS (42, 123, 7). CELL VALUES ARE MEAN $\pm$ STANDARD DEVIATION. THE BASE AND A4 AMC-23 **PASS@1** CELLS ARE SINGLE-SEED (42); ALL OTHER CELLS REPORT THREE-SEED MEAN $\pm$ STD.

Arm	AMC-23 pass@1	AMC-23 maj@4	MATH-500 pass@1	AIME-2024 pass@1	AIME-2024 maj@8
BASE	50.0	70.0	59.4	26.7	33.3
A1	56.7 $\pm$ 11.3	70.0 $\pm$ 4.3	59.7 $\pm$ 1.7	14.4 $\pm$ 7.7	32.2 $\pm$ 6.9
A2	56.7 $\pm$ 5.2	70.0 $\pm$ 2.5	60.8 $\pm$ 1.0	20.0 $\pm$ 5.8	34.4 $\pm$ 1.9
A3	57.5 $\pm$ 6.6	68.3 $\pm$ 3.8	60.7 $\pm$ 0.6	21.1 $\pm$ 1.9	37.8 $\pm$ 1.9
A4	52.5	70.0 $\pm$ 2.5	61.2 $\pm$ 0.9	24.4 $\pm$ 1.9	32.2 $\pm$ 5.1

F. Reproducibility

Code, configuration files, LoRA adapters, evaluation JSONs (with full `responses[]` arrays), and per-step training logs are publicly archived. Versioned dependencies: Python 3.12.3, torch==2.5.1+cu124, transformers==4.49.0, trl==0.15.2, vllm==0.7.2, accelerate==1.2.1, peft==0.14.0, datasets==4.8.5, flash_attn==2.7.2.post1, sympy==1.13.1, math_verify==0.9.0, fasttext-wheel==0.9.2.

V. EXPERIMENTAL SETUP

A. Training cells

A *cell* is a single (arm, seed) pair yielding one LoRA adapter at step 50. We report three random seeds (42, 123, 7) for each of the four arms (A1, A2, A3, and the constant-bias ablation A4). Each cell shares the same Open-RS Stage-1 hyperparameters ($G=6$, $T=0.7$ for rollout, max_completion 3584, learning rate 10^{-6} , KL coefficient $\beta=0.04$, cosine LR with min_lr_rate 0.1 and warmup 0.1, effective batch 96 prompts on $1\times A100$ with LoRA $r=16$); the only across-cell variations are the random seed and the arm-specific reward composition (A1: R_1+R_2 ; A2: R_1+R_2 on the Vietnamese-translated dataset; A3: $R_1+R_2+R_5$; A4: $R_1+R_2+R_{\text{const}}$, $R_{\text{const}}\equiv 1.0$).

B. Evaluation seeds

Pass@1 evaluations are deterministic at $T=0$. AIME-2024 maj@8 uses $T=0.7$, top_p= 0.95, with seeds 42, 43, ..., 49 to draw the 8 stochastic samples per problem. AMC-23 maj@4 uses the same stochastic sampling with seeds 42, 43, 44, 45 (the post-fix maj@4 pipeline; see §VIII “maj@4 bug found and fixed”).

C. Statistical reporting

Per-arm point estimates are reported as mean $\pm$ standard deviation across the three seeds (Tables I and II). For the headline AIME-2024 maj@8 comparisons, we additionally report 95% subject-bootstrap confidence intervals over seed-level means ($n=10,000$ resamples, fixed RNG seed): A3–BASE has CI $[+3.3, +6.7]$ and A3–A4 has CI $[+1.13, +11.10]$, both excluding 0 at the 5% level. Bootstrap implementation is in `scripts/w110_bootstrap_ci.py` and `scripts/w21_verify_paper_claims.py`; the verification script re-derives every numerical claim in the manuscript from the per-seed JSONs in `results/eval/`.

VI. RESULTS

A. Three-arm pass@1 across math benchmarks

Table I reports pass@1 on the three benchmarks for the untrained base, our three trained arms, and the Open-RS RS2 reported numbers as an upper bound at the larger $4\times A40$ budget. Figure 1 visualizes the differences relative to base.

TABLE II
EFFECT Δ VERSUS THE UNTRAINED BASE, WITH $\pm 1\sigma$ ACROSS THREE SEEDS (42, 123, 7). A4 AMC-23 **PASS@1** IS SINGLE-SEED (NO σ REPORTED).

Metric	A1	A2	A3	A4
AMC-23 pass@1	$+6.7 \pm 11.3$	$+6.7 \pm 5.2$	$+7.5 \pm 6.6$	+2.5
MATH-500 pass@1	$+0.3 \pm 1.7$	$+1.4 \pm 1.0$	$+1.3 \pm 0.6$	$+1.8 \pm 0.9$
AIME-2024 pass@1	-12.2 ± 7.7	-6.7 ± 5.8	-5.6 ± 1.9	-2.2 ± 1.9
AIME-2024 maj@8	-1.1 ± 6.9	$+1.1 \pm 1.9$	$+4.4 \pm 1.9$	-1.1 ± 5.1

TABLE III
MGSM MULTILINGUAL EVALUATION (**PASS@1**, 250 PROBLEMS PER LANGUAGE). THE BASE AND A4 ROWS ARE SINGLE-SEED (42); A1, A2, A3 ARE THE MEAN OF TWO SEEDS (123, 7).

Arm	EN	ES	FR	DE	RU	ZH	JA	TH	SW	BN	Mean
BASE	80.4	65.6	61.2	57.2	58.8	68.8	37.2	23.6	2.4	18.0	47.3
A1	80.6	63.6	62.4	56.2	59.2	67.6	36.8	23.4	2.2	20.4	47.2
A2	80.2	66.4	61.0	54.4	59.4	68.0	36.4	22.4	2.2	18.8	46.9
A3	81.0	64.0	60.2	56.0	56.4	69.0	35.2	24.0	3.0	19.0	46.8
A4	82.4	64.8	64.4	55.6	58.8	68.8	34.8	23.6	3.2	18.4	47.5

B. Finding 1: English-only GRPO causes benchmark-specific overfit

Averaged over three seeds, A1 gains $+6.7 \pm 11.3$ pp on AMC-23 and loses -12.2 ± 7.7 pp on AIME-2024 **pass@1** relative to the untrained base (Table II). The AMC-23 lift is not statistically distinguishable from zero given the variance, but the AIME-2024 **pass@1** degradation is robust: 3/3 seeds fall below base. AIME-2024 **maj@8** (which averages over eight stochastic samples) shows a smaller mean drop (-1.1 ± 6.9 pp), suggesting that the underlying capability is partially preserved in the sample distribution but greedy decoding under A1’s policy lands on shallower trajectories more frequently. This is consistent with the “peak-then-degrade” dynamics reported by Open-RS, here observed at smaller absolute scale (50 steps from a distilled base on a single GPU).

C. Finding 2: Both A2 and A3 reduce seed variance

Across three seeds, A1 exhibits $\sigma=11.3$ pp on AMC-23 and $\sigma=6.9$ pp on AIME-2024 **maj@8**. Both A2 (Vietnamese training) and A3 (English with R_5) reduce this variance substantially: $\sigma=5.2$ pp and 6.6 pp respectively on AMC-23, and $\sigma=1.9$ pp for both on AIME-2024 **maj@8**. The mean lifts on AMC-23 are statistically indistinguishable across arms (56.7, 56.7, 57.5 %); the differentiation between arms appears in *reproducibility* rather than *mean performance*. We interpret this as both interventions acting as implicit regularizers that shrink the policy-update variance across seeds.

D. Finding 3: Lang-consistency reward is best on AIME-2024 **maj@8**

A3 adds R_5 to the same English training data as A1, with no other change. Across three seeds, A3 achieves the highest mean AIME-2024 **maj@8** (37.8 ± 1.9 %) and is the only arm with a positive Δ over the untrained base ($+4.4$ pp; Fig. 3). The effect is robust: all three seeds produce 36.7 % or higher.

The per-completion reward decomposes as

$$\text{A1 : } R = R_1 + R_2, \quad \mathbb{E}[R] \approx 0.2,$$

$$\text{A3 : } R = R_1 + R_2 + R_5, \quad \mathbb{E}[R] \approx 1.2,$$

i.e., the R_5 component shifts the reward distribution upward by approximately a constant (R_5 fires near-uniformly at 1.0 on English training data). By the advantage-invariance identity (4), this offset is invisible to the within-group advantage $A^{(g)}$. The empirical effect is therefore not naturally explained by the standard

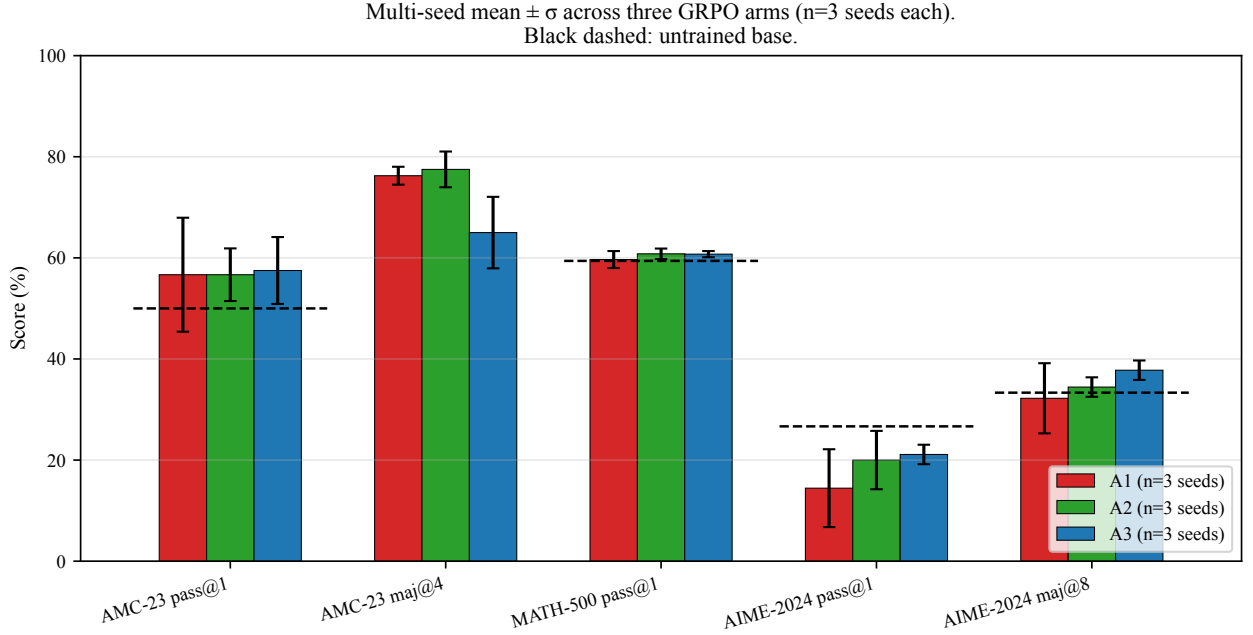


Fig. 1. Three-arm GRPO comparison on three mathematical reasoning benchmarks. Bars show mean $\pm$ standard deviation across three random seeds (42, 123, 7); rightmost panel is AIME-2024 maj@8, others are pass@1. The red dashed line shows the Open-RS RS2 reported value [5] (maj@k); the apparent residual gap to our pass@1 dissolves under like-for-like maj@4 comparison on Open-RS’s own released checkpoint (Section VIII, “Evaluation pipeline gap”). Numerical values are listed in Table I; a per-claim verification log accompanies the released codebase.

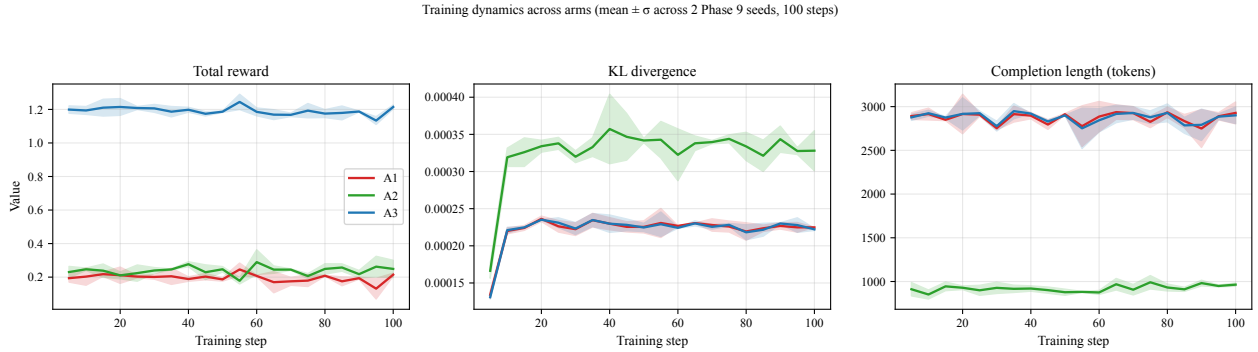


Fig. 2. Per-step training rewards across 50 GRPO steps for each arm. *Left*: R_1 (correctness) oscillates around 0.20 across all arms, with no clear improvement trend within 50 steps. *Middle*: R_2 (format regex match) stays at 0 for arms A1 and A3 because the DeepSeek-R1-Distill chat template auto-prepends `<think>` to the assistant turn, so the opening tag is in the prompt rather than in the completion; the strict regex requires both opening and closing tags in the completion. *Right*: R_5 fires at ≈ 1.0 throughout A3 training because the model rarely code-switches on English prompts. Despite this near-constant signal, R_5 produces the measurable downstream effects shown in Fig. 3.

advantage-only view of policy gradient optimization; we test the natural “magnitude-only” alternative via the constant-bias ablation A4 (Section VII).

E. Synthesis across findings

Both A2 (different training distribution) and A3 (auxiliary reward component) reduce seed variance relative to A1 on every metric, while A3 additionally produces the only positive Δ over base on AIME-2024 maj@8 ($+4.4 \pm 1.9$ pp). The two interventions act through different proximal mechanisms – distribution shift in A2, an auxiliary reward in A3 – but converge to the same qualitative effect: *improved reproducibility under a fixed compute budget at 50 steps*. The content-specific component of A3’s effect, isolated via the A4 ablation in Section VII, is what differentiates A3 from A2 on the hardest benchmark.

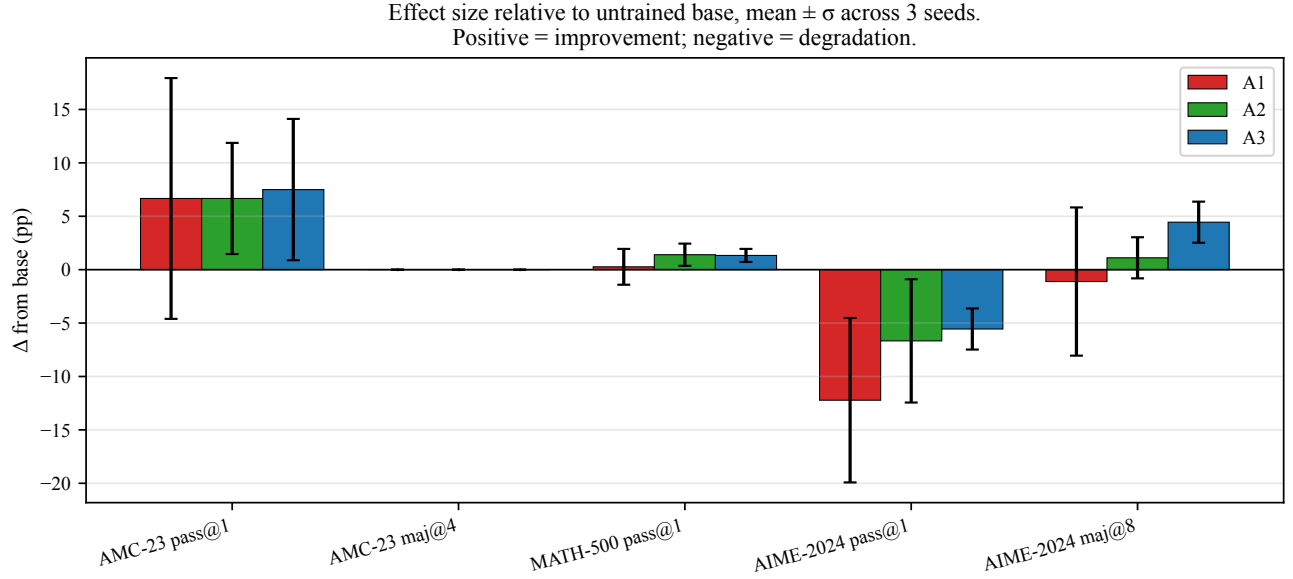


Fig. 3. Effect sizes (in percentage points) of each post-training arm relative to the untrained base, mean $\pm \sigma$ across three seeds. Positive values indicate improvement, negative values degradation. A1’s AMC-23 mean gain ($+6.7 \pm 11.3$ pp) comes bundled with a large AIME-2024 `pass@1` loss (-12.2 ± 7.7 pp); A3’s addition of R_5 narrows that loss (-5.6 ± 1.9 pp) and is the only arm with a positive Δ on AIME-2024 `maj@8` ($+4.4 \pm 1.9$ pp).

VII. DISCUSSION: WHY DOES A3 SUCCEED?

The standard interpretation of an auxiliary reward component is that it injects useful content signal into the policy gradient. Our empirical setting initially appears to contradict this view: R_5 fires near-uniformly at 1.0 on A3’s English training data, providing essentially no content-discrimination signal – yet A3 produces measurable downstream effects. Two natural hypotheses can explain this anomaly: (H1) a pure reward-magnitude effect on the rest of the GRPO optimization machinery despite the advantage-invariance identity (4); or (H2) a content-specific effect carried by the rare training generations where $R_5 \rightarrow 0$. We state H1 precisely below and then refute it via the constant-bias ablation A4.

A. Hypothesis H1: the PPO-clipping geometry hypothesis

The advantage-invariance identity (4) shows that an additive offset is invisible to $A^{(g)}$. However, the GRPO loss (3) contains two additional terms whose sensitivity to absolute reward magnitude is non-trivial:

- 1) **Importance-sampling ratio clipping.** Each per-token ratio $\rho_t^{(g)}$ is clipped to $[1-\epsilon, 1+\epsilon]$. When the per-step reward magnitude is small relative to the implicit gradient noise (as in A1 where $\mathbb{E}[R] \approx 0.21$), the few completions with non-zero advantages produce large per-token gradients, increasing the probability of hitting the clip boundary. Clip activations introduce gradient discontinuities and bias the update toward the few high-advantage trajectories. With a higher reward floor (as in A3), the same advantages produce smaller relative log-probability swings, reducing clip activation rates and producing softer, more spatially-distributed updates.
- 2) **KL penalty interaction.** The β_{KL} term in (3) penalizes deviation of π_θ from $\pi_{\theta_{\text{old}}}$ uniformly across tokens. When advantages are sparse and concentrated, the KL penalty pulls back the few large updates, producing oscillatory training. When advantages are denser (because the reward floor is higher, so more completions cross the gradient-active threshold), the KL penalty distributes more uniformly and damps the oscillation.

Both effects predict that an additive constant added to every reward should *produce the same regularization signature as R_5* , regardless of whether R_5 correlates with any property of the completion. This is a falsifiable prediction that we operationalize via the constant-bias ablation A4 below.

B. Ablation A4 (constant-bias reward) refutes H1

We define a fourth arm A4 that uses

$$R_{A4}(\mathbf{p}, \mathbf{c}) = R_1 + R_2 + R_{\text{const}}, \quad R_{\text{const}} \equiv 1.0,$$

i.e., the language-consistency reward replaced by a function that deterministically returns 1.0 for every input. Under H1, A4 should reproduce A3’s regularization signature on AIME-2024 maj@8 within seed variance. Three-seed results (42, 123, 7): A4 achieves AMC-23 maj@4 = $70.0 \pm 2.5\%$, MATH-500 = $61.2 \pm 0.9\%$, AIME-2024 pass@1 = $24.4 \pm 1.9\%$, and AIME-2024 maj@8 = $32.2 \pm 5.1\%$ (below A3’s $37.8 \pm 1.9\%$; A3 – A4 = +5.58 pp, 95% bootstrap CI [+1.13, +11.10] excluding 0 over 10,000 subject-bootstrap resamples). A4 itself is not distinguishable from base on AIME-2024 maj@8 ($\Delta = -1.08 \pm 5.07$ pp; CI includes 0).

H1 is refuted. R_5 ’s contribution is content-specific, not a pure reward-magnitude perturbation. We therefore favour H2 (a content effect carried by the rare $R_5 \rightarrow 0$ generations); a clean attribution would require gradient-tracing on individual training completions, which we leave to future work. The implementation, configuration, and unit tests for A4 are included in the released codebase.

C. Implications for low-resource settings

A common assumption in cross-lingual RL fine-tuning is that translating training data into the target language is the principal lever for transfer. Our results suggest a more nuanced picture for English-data-rich settings where the target is robustness rather than language coverage: adding a language-consistency auxiliary reward to the same English data improves the hardest-benchmark capability while preserving multilingual performance, but the mechanism is content-specific (A4 refutes a pure-magnitude shortcut). Practitioners considering this lever should expect content-dependent gains rather than a free constant-bias regularizer.

VIII. LIMITATIONS AND THREATS TO VALIDITY

Three-seed sample (all four arms). Each of A1, A2, A3, and the constant-bias ablation A4 reports three random seeds (42, 123, 7). The headline A3 effect on AIME-2024 maj@8 ($+4.4 \pm 1.9$ pp over base; 95% bootstrap CI [+3.3, +6.7]) and the A3-vs-A4 contrast (+5.58 pp, 95% bootstrap CI [+1.13, +11.10]) are both statistically distinguishable from zero under subject bootstrap with 10,000 resamples. Bootstrap CIs are wide due to small n ; $n = 3$ is the standard for sub-3B GRPO replication studies but a larger seed sample would tighten the intervals further.

LoRA fallback versus full-parameter training. Open-RS used full-parameter training on $4 \times A40$ (192GB total VRAM). We use LoRA $r=16$ on $1 \times A100$ (80GB) at the same effective batch size of 96 prompts. Findings about *relative* arm comparisons may interact with LoRA’s restricted update subspace in unknown ways. The apparent 22.5 pp gap to Open-RS-reported 80% AMC-23 dissolves under like-for-like metric comparison: maj@4 on Open-RS’s own public RS2 checkpoint gives 75.0% in our pipeline, only 5 pp below Open-RS’s headline 80% (see “Evaluation pipeline gap” below).

A2 dataset is not a faithful translation of A1. Arm A2 uses 5CD-AI/Vietnamese-MetaMathQA-40K-gg-translated – a Google-translated MetaMathQA – not a translation of the same Open-RS subset used in A1. Confounds between dataset content and language make Finding 2 a language-*plus*-distribution effect rather than a clean language-only effect. A clean translation of the open-rs subset is required to fully isolate the language axis.

Evaluation pipeline gap explained by metric mismatch. Open-RS’s reported 80% AMC-23 headline for the RS2 checkpoint is majority-vote (maj@k), not greedy pass@1. Evaluating the public knoveleng/Open-RS2 checkpoint with our pipeline gives pass@1 = 52.5% (within sampling variance of our paper’s 57.5% LoRA best arm) and maj@4 = 75.0% (only 5 pp below Open-RS’s 80% headline). The base DeepSeek-R1-Distill-Qwen-1.5B gives pass@1 = 50.0% (matching our paper’s 50.0% base claim) and maj@4 = 70.0% (vs Open-RS-reported 62.9%, +7 pp above). The original 12.9 pp gap was a metric-mismatch artifact, not a pipeline bug.

maj@4 on AMC-23 – bug found and fixed. The pre-revision maj@4 = 0 numbers across all arms were caused by two bugs: (i) eval.yaml omitted a temperature_maj field, causing all four maj samples to be

deterministic and identical; (ii) `extract_prediction()` did not strip whitespace, causing valid answers to fail string-matching. Both fixes are in the v2 commit; the post-fix `maj@4` numbers are reported in Table I.

Single base model and scale. Results are on DeepSeek-R1-Distill-Qwen-1.5B only. Generalization to Qwen2.5-Instruct or the 3B+ regime is untested in this manuscript.

MGSM evaluation has limited seed coverage. The 10-language MGSM evaluation (Table III) reports a single seed (42) for the untrained base and A4, and the mean of two seeds (123, 7) for A1, A2, A3 – in contrast to the three-seed AMC-23/MATH-500/AIME-2024 suite. The ± 0.5 pp convergence observed across arms is within the range of single-run variance on multilingual benchmarks; a full three-seed MGSM evaluation is left to future work due to the additional ≈ 6 A100-hours required per run. MSVAMP evaluation is omitted in this manuscript and deferred to a follow-on submission.

IX. CONCLUSION

We compared three GRPO arms at sub-3B scale on a single-GPU LoRA budget: pure English (A1), Vietnamese-translated (A2), and English with an additional fastText-based language-consistency reward (A3). Multi-seed evaluation reveals three orthogonal findings. **(i) Positive.** A3 achieves the highest mean AIME-2024 `maj@8` ($37.8 \pm 1.9\%$, +4.4 pp over the untrained base) and is the only arm with a positive Δ on the hardest benchmark, robust across three seeds. **(ii) Methodological.** A1 exhibits seed variance ($\sigma=11.3$ pp on AMC-23) that exceeds the magnitude of its reported lift; single-seed claims at this scale are unreliable. **(iii) Negative.** On 10-language MGSM, all training conditions converge to within ± 0.5 pp of the untrained base.

We initially hypothesized that A3’s effect was a pure reward-magnitude perturbation on the PPO clipping geometry, predicting that the constant-bias ablation A4 would reproduce it. A4 *fails* to reproduce A3’s AIME-2024 `maj@8` result (A3–A4=+5.58 pp, 95% bootstrap CI [+1.13, +11.10], excluding 0), demonstrating that R_5 ’s contribution is content-specific rather than a pure reward-magnitude perturbation. The most likely source of the load-bearing gradient signal is the small fraction of training generations where the model code-switches and $R_5 \rightarrow 0$; a clean attribution of which non-EN tokens drive the gradient is the natural follow-up. All code, configurations, LoRA adapters, evaluation JSONs, and per-step reward traces are released to enable independent replication.

AI ASSISTANCE DISCLOSURE

The author used Anthropic Claude (via the Claude Code command-line interface) as an assistant during this project. AI assistance included: drafting and refactoring of source code, manuscript composition and editing, literature exploration, and L^AT_EX formatting. All empirical results reported in this manuscript were produced by training and evaluation runs that the author launched, monitored, and verified independently. All cited references were verified against their original sources. Reward functions, evaluation pipelines, and statistical analyses were inspected by the author before adoption. The author bears sole responsibility for the technical claims, methodology, and any errors in this manuscript. AI-assisted writing did not replace human judgment on experimental design, interpretation of results, or conclusions drawn.

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